1. Information about the indicator	
Goal	Ensuring the availability and rational use of water resources and sanitation for all
Task	By 2030, significantly improve the efficiency of water use in all sectors and ensure sustainable intake and supply of fresh water to address water scarcity and significantly reduce the number of people suffering from water scarcity.
The indicator	Dynamics of changes in water use efficiency by type of economic activity
2. Information about the organization	
Organization	Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan
3. Definitions and concepts	
Definition	Changing the efficiency of water use over time. The change in water use efficiency is expressed as a percentage of the water use efficiency of the reporting year compared to the previous one.
Basic concepts:	The efficiency of water use is determined by the ratio of the volume of gross value added to the total volume of water used. Gross value added is the difference between the output of goods and services and the cost of products fully consumed during production (intermediate consumption).
Justification and interpretation	The purpose of this indicator is to provide information on the effectiveness of the economic and social use of water resources in the main sectors of the economy. The indicator provides signals for improving water use efficiency in all sectors, highlighting those sectors where water use efficiency lags behind. The calculation is carried out in accordance with the AQUASTAT methodology of the Food and Agriculture Organization of the United Nations (FAO).
4. Data sources and collection methods	
Data sources	The data source for the indicator formation is the data from the departmental statistical observation "Report on water intake, use and sanitation" (index 2-TP (water farm), frequency - annual). Data on water intake and use is recorded by the Committee for Regulation, Protection and Use of Water Resources of the Ministry of Water Resources and Irrigation of the Republic of Kazakhstan. GVA data for individual sectors of the economy at constant prices in tenge and in US dollars at 2015 prices - official statistics data, FAO.
Data collection methods	Data collection is based on a comprehensive survey of water users who use water for agricultural, industrial, municipal, and hydropower needs.
Unit of measure ment	thousand tenge/ m3, US dollars / m3
Periodicity	annual
5. Calculation method and other methodological considerations	
Calculation method:	Water use efficiency is calculated for 3 sectors of the economy (agriculture (Section A), industry (Sections B, C, D, E), and other activities (sections G-T)). Collectively, the three sectoral efficiencies provide an indicator of the overall efficiency of water use in the country. For each sector of the economy, the indicator is calculated as the ratio of gross value added to the volume of water withdrawn in

	this industry over a certain period of time. To calculate the aggregated value for the entire economy, the indicator is calculated as the sum of the three sectors listed above, weighted according to the proportion of water taken by each sector. In the formula: $$WUE = Awe \times P_A + Mwe \times P_M + Swe \times P_S$$, where: WUE - water use efficiency (USD/m3), Awe - efficiency of water use in irrigated agriculture (USD/m3), Mwe - industrial water efficiency (USD/m3), Swe - efficiency of water use in other activities (USD/m3), P_A, P_M, P_S – the share of water taken in the agricultural sector, industry, and other activities in the total volume of water intake, respectively. The efficiency of water use in irrigated agriculture is calculated as the ratio of gross value added in agriculture to water intake in the sector, expressed in dollars/m3. In the formula: $$Awe = \frac{GVAa \times (1 - Cr)}{Va}$$,where: GVAa - gross value added of agriculture (excluding fishing and forestry), dollar Cr – the share of GVA produced in non-irrigated agriculture, Va - volume of water intake in agriculture, m3. Cr It is calculated based on the share of irrigated land in the total area of cultivated land, namely: $$Cr = \frac{1}{1 + \frac{Ai}{(1 - Ai) * d}}$$ where: Ai - the share of irrigated land in the total area of cultivated land, with a decimal place, d - the ratio between rain-fed and irrigated crops (established by FAO, in 2017-2018, d=0.563). The calculation of the efficiency of industrial water use in other types of activities is calculated as the ratio of GVA to the volume of withdrawn water in the sector.
Comments and restrictions:	
6. Data availability and disaggregation	
Data availability and gaps:	The data is available on the official website of the Bureau of National Statistics. Time series: since 2015.
The level of disaggregation:	in general, for the entire economy, at the level of the main sectors of the economy: - Agriculture (section A), - Industry (Sections B, C, D, E), - Other activities (sections G-T)
7. Comparability with international	

data/standards	
8. References and documentation	